

31338 Network Servers

32520 Systems Administration

Week 6

Domain Name System (DNS)

Dr Litao Yu

University of Technology, Sydney

Domain Name System (DNS)

- Recall from last time: **When will we use this service?**
 - IP address – numeric label of a host
- Domain names : two parts, **the host name** and **the domain name**
 - Assign IP **addresses** into domain names, e.g., **www.uts.edu.au**
 - FQDN: A Fully Qualified Domain Name for a specific host: **the hostname.domainname => e.g., mymail.uts.edu.au**
 - Domain name can be hosted anywhere with the **changes of IP**
 - So, keep the same domain name (to keep customers), but IP can be changed with your web hosting service
- ICANN - **I**nternet **C**orporation for **A**ssigned **N**ames and **N**umbers
 - Maintenance and procedures of DNS and manage IPv4/v6
 - The **IANA*** is the part of ICANN for managing the DN, IP etc.
 - The IANA manages the registries of DN, protocol parameters and IP address

Before Domain Name System (DNS)

- Centrally held list of host names and IP addresses
 - Called HOSTS.TXT
 - Downloaded periodically by Internet hosts
 - Not a scalable solution
- Modern equivalent lives on in Unix and Windows
 - `/etc/hosts`
 - `C:\Windows\System32\drivers\etc\hosts` (Windows)
 - Lists IP addresses and host names (can include FQDN*)

IP Address:	Host Name	Alias
127.0.0.1	localhost.localdomain	localhost
192.168.1.1	fang.cats.org	fang
192.168.1.2	moggy.cats.org	moggy

DNS Name Space: hierarchal structure

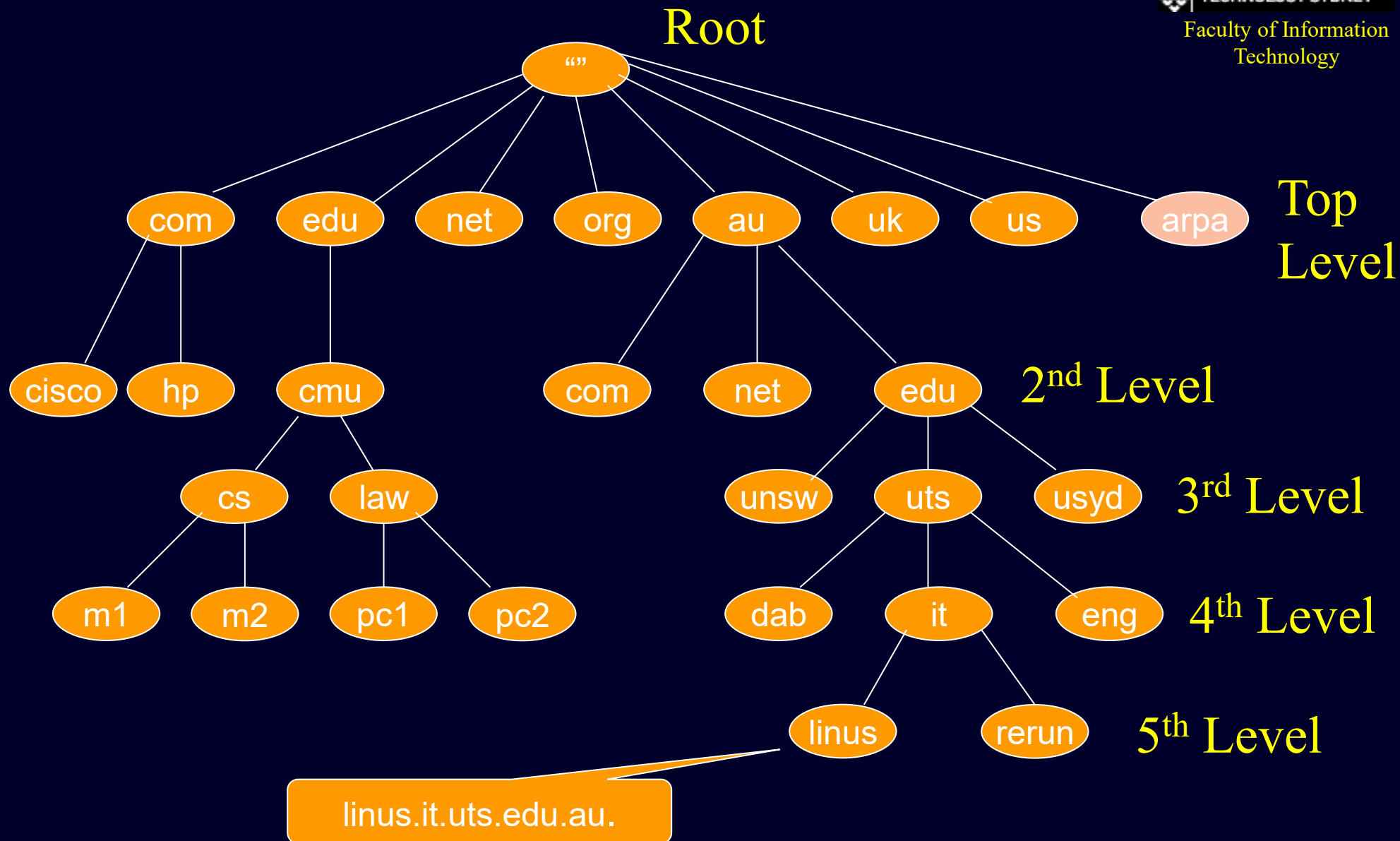


UNIVERSITY OF
TECHNOLOGY SYDNEY

Faculty of Information
Technology

- Upside-down tree structure ()
 - A web host is regarded by hostname and domain, separated by dots, e.g. www.uts.edu.au
- Root Domain (null): represented by "" or . (by 12 diff org.*)
- Top Level Domains (TLD) (First-Level)
 - Generic: .org, . gov, .com, .edu, and now .biz, .info, .tv, etc
 - Country code: e.g., .au, .uk, .us, .md, .nz, etc
 - ICANN selects organizations to manage TLD
- Second-Level domains (and further)
 - 2nd level domain (**edu.au** & **ns.edu.au**)
 - 3rd level domain (**uts.edu.au** & **ns.uts.edu.au**)
 - 4th level domain (**it.uts.edu.au** & **ns.it.uts.edu.au**)

* The root DNS servers are operated by 12 different organizations: including Verisign, US Department of defense, US army research lab, University of Southern California, ICANN, RIPE, WIDE, Internet Systems Consortium



Basic Concept of DNS

- DNS (~Lookup table, the IP address for a domain...) is for:
 1. resolving domain to IP addresses (IPv4 **A** records, IPv6 AAAA records)
 2. host aliases (**CNAME** records) #Canonical name record (**subdomains**)
 3. specifying name server(s) for a domain (**NS** records)
 4. specifying mail server(s) for a domain (**MX** records) #Mail eXchange
 5. some others less often used (**TXT**, **SRV**, etc)
 6. stores admin information about a domain (Start of Authority **SOA** records)
 7. IP addresses to domain (reverse lookup, **PTR** records) #PoinTeR
 - see `/etc/named.conf`

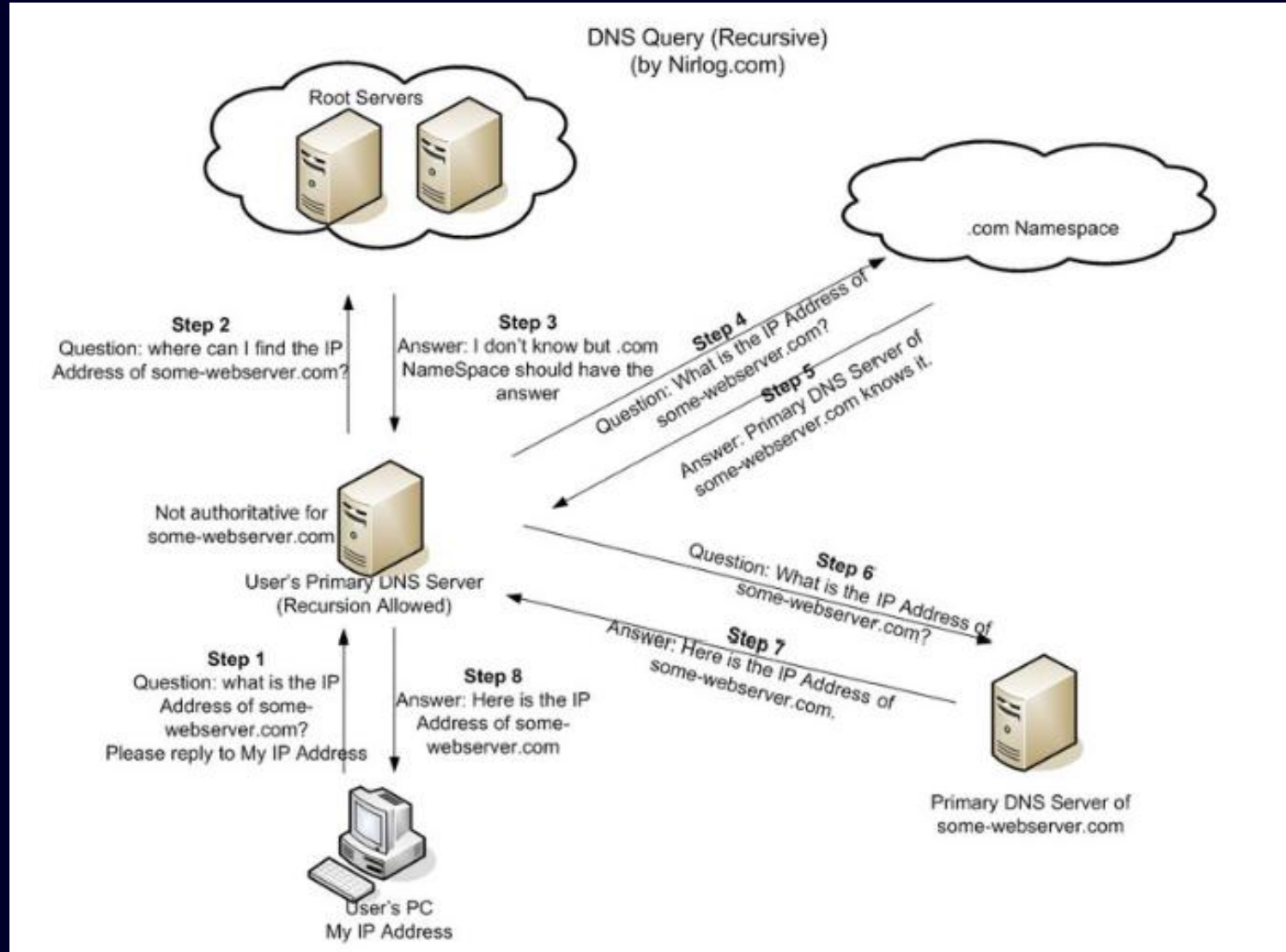
DNS slaves and caching

Roughly defined terminology and to be configured as:

- Primary (Master) DNS (**only one**):
 - Required, **authoritative** server
- Secondary (Slave) DNS (**multiple and redundancy**):
 - Optional, to improve performance
 - periodically downloads/copy a zone file (**read only**) from a master DNS server: **Zone is a portion of the DNS.**
 - 2nd DNS can then answer authoritative queries about that zone
- Caching-only DNS (mainly used by ISP) **Interment Service Provider**
 - gather a large cache of DNS responses to improve DNS response times significantly (DNS performance) – **Fast!**
 - Time-To-Live (**TTL***) before it must download the data again

* A TTL stands for the amount of time data should remain valid and available before a computing system discards it

Anatomy of a full DNS search



Q&A

The .edu domain is considered which domain type? (Choose the best answer.)

- A. Root domain
- B. TLD
- C. ICANN
- D. Second-level domain

Based on the last slide, a program or library routine that creates a DNS query, checks its own cache for the answer, and if it doesn't find it there, sends the query to another name server, is called what? (Choose the best answer.)

- A. Name server
- B. Cache
- C. Name resolution
- D. Resolver
- E. Zone

When you type a domain name into your web browser, the DNS resolver takes that domain name and translates it into the corresponding internet protocol (IP) address. It does this by querying a series of DNS servers until it finds the IP address associated with the domain name. -- **Reslover**

DNS and Bind

- How to set up a DNS server? Install a BIND S/W package
 - **Berkeley Internet Name Daemon (bind)**, an implementation
 - `yum/dnf install bind`
 - **DNS Server is called** named daemon ("name-dee")

For queries:

- command-line interfaces to DNS (**dig**, **host**, **nslookup**),
#dig: A DNS lookup tool (short for Domain Information Groper). It's commonly used on Linux/Unix systems to query DNS records.
#dig replaces older tools such as nslookup and host

nslookup **www.uts.edu.au** #use the first name server in /etc/resolv.conf

dig **uts.edu.au** **ns** #

dig **www.uts.edu.au** **a** # **a**: Specifies the query type. The **A record** maps a domain name

dig **@ns.uts.edu.au** **www.uts.edu.au** **any** # Asks the UTS nameserver

(ns.uts.edu.au) to return all known DNS records for www.uts.edu.au.

- Other implementations: djbdns, dnsmasq, pdnsd, PowerDNS

Bind configuration – Server

Step1: Edit `/etc/named.conf` (make a copy for backup first!)

– clauses (statements) to control operation for each zone

It identifies the zone name and its record file

```
zone "it.netserv.edu.au" {  
    type master; #slave, forward, hint  
    file "it.netserv.edu.au.zone";  
    #configuration file  
};
```

Step2: Create `/var/named/it.netserv.edu.au.zone`

- This is the "zone file" for the zone `it.netserv.edu.au.zone`
- Copy it from `/var/named/named.empty` (make a copy!)
- make sure the file permissions are OK

Do you know `chmod/chown/chgrp`? (mod, ownership, groupship?)

Bind configuration – Forward lookup

Zone file: /var/named/it.netserv.edu.au.zone

\$TTL 3H

name	class	type	record specific information; @=ORIGIN
@	IN	SOA	ns.it.netserv.edu.au. root@it.netserv.edu.au. (
			20 ; serial
			1D ; refresh
			1H ; retry
			1W ; expire
			3H) ; minimum
	IN	NS	ns.it.netserv.edu.au
	IN	MX	10 mail
	IN	A	10.0.2.1
ns	IN	A	10.0.2.1
site	IN	A	10.0.2.10
www	IN	CNAME	site
ftp	IN	CNAME	site
mail	IN	CNAME	10.0.2.10

#www is an alias name; 10 before mail is <preference-value> allows you to numerically rank the email servers. The lowest or the highest preferred.

Record Class (IN): The IN (Internet) class will be used for the Managed DNS service

Bind configuration– Reverse lookup

Zone file: /var/named/2.0.10.in-addr.arpa.zone

Step1: /etc/named.conf

```
zone "2.0.10.in-addr.arpa" {  
    type master;  
    file "2.0.10.in-addr.arpa.zone";};
```

Add another arpa(rev)zoon in the conf file !

Step2: /var/named/2.0.10.in-addr.arpa.zone

\$TTL 3H

;name class type record specific information; @=\$ORIGIN

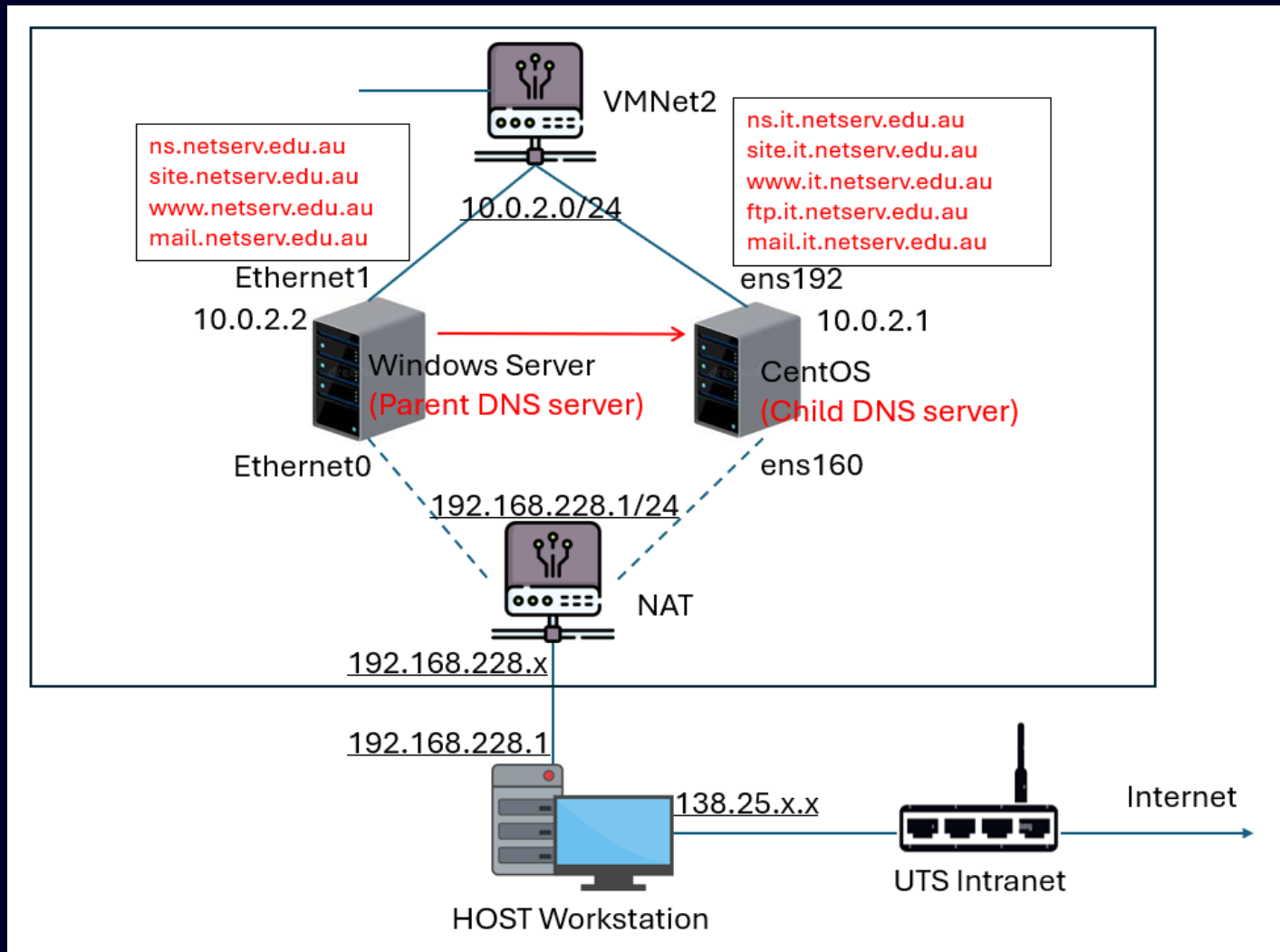
@	IN	SOA	ns.it.netserv.edu.au. root@it.netserv.edu.au (
			20 ; serial
			1D ; refresh
			1H ; retry
			1W ; expire
			3H) ; min negative cache
10	IN	NS	ns.it.netserv.edu.au.
	IN	PTR	site.it.netserv.edu.au.

The "@" symbol is a special label that indicates the \$ORIGIN should replace the "@" symbol. This is typically used for the apex of a zone. SOA Record – The \$ORIGIN is followed by the zone's Start Of Authority (SOA) record. An SOA record is required for each zone

Lab 6 Configuration

DNS name	IP address	Type	Server	Notes
netserv.edu.au	10.0.2.2	NS	Windows	Primary standard zone
ns.netserv.edu.au	10.0.2.2	A	Windows	Actual Primary DNS
mail.netserv.edu.au	10.0.2.20	A	Windows	Actual Mail server
site.netserv.edu.au	10.0.2.20	A	Windows	Actual Web server
www.netserv.edu.au	10.0.2.20	CNAME	Windows	Alias to site.netserv.edu.au
it.netserv.edu.au	10.0.2.1	NS	CentOS	Secondary standard zone
ns.it.netserv.edu.au	10.0.2.1	A	CentOS	Actual Primary DNS
site.it.netserv.edu.au	10.0.2.10	A	CentOS	Actual ftp and www server
www.it.netserv.edu.au	10.0.2.10	CNAME	CentOS	Alias to site.it.netserv.edu.au
ftp.it.netserv.edu.au	10.0.2.10	CNAME	CentOS	Alias to site.it.netserv.edu.au
mail.it.netserv.edu.au	10.0.2.10	A	CentOS	Actual Mail server

Lab 6a: DNS server configuration



Client (to identify where is the DNS Server)

- DNS client

Generated by NetworkManager!

when you open a browser and type a domain, your computer's DNS client asks a DNS resolver for the IP. (/etc/resolv.conf)

```
search it.uts.edu.au uts.edu.au #default search list, when
resolving short hostnames such as test, it will search
test.it.uts.edu.au, then test.uts.edu.au
```

```
nameserver 10.0.2.2 (DNS server1 IP address-Windows)
```

```
nameserver 10.0.2.1 (DNS server2 IP address-Linux)
```

- resolve.conf can be set by DHCP (lab 4, Networking Skills Test!)
- resolve order determined by /etc/nsswitch.conf

```
hosts:          files dns myhostname # /etc/hosts first, then dns, then `myhostname`
```

```
[root@fff ~]# cat /etc/hosts
IPv4-only 127.0.0.1 localhost localhost.localdomain localhost4
           localhost4.localdomain4
IPv6-only ::1      localhost localhost.localdomain localhost6 localhost6.localdomain6
```

- test in Lab with **dig**, by **not** using the **@localhost** option
 - i.e., make it use the default server (from the resolv.conf file)

Client Configuration

- Change the resolve order
 - Manual changes in `/etc/resolv.conf`, **Not recommend!**
 - *Disable auto-config: modify `/etc/NetworkManager/NetworkManager.conf` and add "dns=none" under [main] section*
 - *`systemctl restart NetworkManager`*
 - *manual update `/etc/resolv.conf`, and add "nameserver 10.0.2.2" "nameserver 10.0.2.3, ..."*
- **nmcli** fast way but not in favour!
 - `nmcli con mod ens192 ipv4.dns "10.0.2.1 10.0.2.2"`
 - `nmcli con reload ens192`
 - `nmcli con up ens192`
 - `man NetworkManager.conf`, `man nmcli` #check manual

Client Configuration – NetworkManager

This way is recommended!

Add the following parameters:

- Client config is separate for each interface in
`/etc/NetworkManager/system-connections/ens192.nmconnection`

```
dns=10.0.2.2;10.0.2.1
```

```
dns-search=netserv.edu.au;it.netserv.edu.au;
```

```
dns-priority=10
```

The search domains for all interfaces are **joined together** in a single search line in
`/etc/resolv.conf`

- The nameservers for each interface are added to `resolv.conf`
- The order of search domains and name servers in `resolv.conf` depends on the DNS priority value of the interface (and interface number when priority is the same) -- `/etc/nsswitch.conf`
- Previous manual changes to `resolv.conf` will be overwritten